

Indian Statistical Institute

BSDS II

Stochastic Process

Class Test II

Time : 1 hour 15 Minutes

Total marks 17. Maximum you can score is 10.

Attempt all questions. Statements like “Clearly,” or “This is easy to see” are **strongly discouraged**.

1. Consider the Markov Chain on $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$ with the transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 & 0 \\ \frac{2}{5} & \frac{1}{3} & 0 & 0 & 0 & 0 \\ 0 & \frac{1}{5} & \frac{1}{6} & \frac{1}{3} & 0 & \frac{1}{6} \\ \frac{1}{4} & \frac{1}{3} & \frac{1}{6} & \frac{1}{8} & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & 0 & \frac{1}{8} & 0 \end{bmatrix} \end{matrix}.$$

Find the values of f_{33}^* and f_{44}^* . What is the probability that the chain starting from 3, visits the state 4 before visiting state 6. [3+3+4]

2. Consider the Markov Chain on $\mathcal{S} = \{1, 2, 3, 4\}$ with the transition matrix $\mathbf{P}$ given by

$$\mathbf{P} = \begin{array}{c} \begin{array}{cccc} & 1 & 2 & 3 & 4 \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ \frac{2}{5} & \frac{3}{5} & 0 & 0 \\ 0 & \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{array} \end{array}.$$

Find all the stationary distributions.

[7]

Rough page

Rough page